Alzheimer's Research & Therapy

RESEARCH

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Automated high-throughput quantification of plasma p-tau217 and APOE-ε4 for Alzheimer's disease diagnosis and cognitive decline in a memory cohort

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Abstract

Background Blood-based biomarkers could improve the precision of Alzheimer's disease (AD) clinical diagnosis and expand access to targeted treatments. Therefore, we evaluated the diagnostic accuracy of plasma Elecsys p-tau217 (Roche) and compared it with Elecsys p-tau181 (Roche) and Lumipulse p-tau217 (Fujirebio). We also assessed the added value of APOE-ε4 carrier status, plasma Aβ42 and Aβ42/40 in a memory-clinic cohort and evaluated associations with longitudinal cognition.

Materials and methods A total of 187 patients from the Cognitive Centre Ghent University (CCUG) biobank, classified as AD ($n=103$) or non-AD cognitive disorders ($n=84$) based on CSF biomarkers (CSF Aβ42/40 ratio, total tau and p-tau181), were included. Plasma Elecsys p-tau181, p-tau217, and APOE-ε4 were measured on the Roche cobas® pro e801, and p-tau217, Aβ42, and Aβ40 on the Fujirebio LUMIPULSE G1200. ROC analyses and single- and two-cut-off strategies (95% sensitivity/specificity) were applied. Subgroup analyses examined APOE-ε4 status, renal function, cerebral amyloid angiopathy (CAA) features, Fazekas score, and longitudinal cognition.

Results Elecsys plasma p-tau217 showed high discriminative performance for AD versus non-AD (AUC 0.939), comparable to Lumipulse p-tau217 (AUC 0.950; $p=0.485$). Elecsys p-tau181 performed lower than Elecsys p-tau217 (AUC 0.903; $p=0.043$). Using a two-cut-off strategy, the intermediate proportion was 19.9% for Elecsys p-tau217, 11.9% for Lumipulse p-tau217, and 33.2% for Elecsys p-tau181. Adding APOE-ε4 to Elecsys p-tau217 improved discriminative performance (AUC 0.970, $p=0.02$) and reduced intermediates to 11.0%. Adjustment for Aβ42 on the Fujirebio platform did not significantly increase the AUC (0.950 vs. 0.957; $p=0.322$) and modestly reduced intermediate classifications (11.9% to 10.0%). Higher baseline Elecsys p-tau217 was associated with lower baseline MoCA and a trend towards faster MoCA decline ($p=0.07$). Age, sex, renal function, Fazekas score, and CAA were not significantly associated with Elecsys p-tau217 concentrations.

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Conclusion Plasma Elecsys p-tau217 measured on an automated high-throughput platform shows excellent diagnostic accuracy for AD. Incorporating APOE-ε4 further improves classification, while Aβ42 adjustment had only limited additional impact. Baseline p-tau217 also reflects cognitive severity and may relate to subsequent cognitive decline in the memory-clinic setting.

Keywords Alzheimer's disease, blood biomarkers, plasma p-tau217, diagnostic performance, cognitive decline

Background

Blood biomarkers for Alzheimer's disease (AD) have advanced rapidly over the past decade and are now increasingly implemented in research, clinical trials, and, more recently, routine clinical practice [1, 2]. Compared with cerebrospinal fluid (CSF) analysis and positron emission tomography (PET) imaging, blood-based measures are less invasive, more accessible, less costly, and generally more acceptable to patients, making them particularly attractive for large-scale application in memory clinics [3].

Recent literature highlights the value of plasma biomarkers such as phosphorylated tau (p-tau181, p-tau217), amyloid-β42/40 ratio, neurofilament light chain (NfL), and glial fibrillary acidic protein (GFAP) [4–12]. Among these, plasma p-tau217 has consistently emerged as the most accurate blood marker for AD pathology, with multiple studies demonstrating better discriminative ability than p-tau181 [13–15], whereas plasma NfL reflects neurodegeneration and plasma GFAP has been associated with astrocytic injury and reactive astrogliosis. Reported areas under the receiver operating characteristic (ROC) curve for p-tau217 typically exceed 0.90 for distinguishing AD from other non-AD cognitive disorders, approaching the diagnostic accuracy of CSF and PET biomarkers [14].

However, most published studies of p-tau217 have relied on research-use assays (e.g., Fujirebio, Quanterix, ALZPath) that require batch processing and are less suited for large-scale routine use, limiting their applicability in clinical practice [14]. Automated, high-throughput platforms are essential for implementation in routine practice. The Roche Elecsys system is already widely available in clinical laboratories and provides fully automated, high-throughput immunoassays. Elecsys plasma p-tau181 has been evaluated in several clinical cohorts, showing moderate-to-high accuracy for Aβ-PET and CSF positivity (AUCs 0.80–0.89), with further improvement when combined with Elecsys Aβ42/40 or in multiparametric panels (AUCs up to 0.93) [16–19]. Very recently, in October 2025, the FDA cleared the Elecsys Phospho-Tau 181P (p-tau181) assay as a rule-out test (negative predictive value 97.9%) for adults aged ≥ 55 years presenting with complaints of cognitive decline in primary care. By contrast, data on plasma p-tau217 on the Elecsys platform, although potentially more accurate, remain scarce, and establishing its diagnostic performance is a key next step [20].

Several factors may influence plasma biomarker levels and their interpretation. The incremental value of APOE-ε4 testing remains uncertain, with some cohorts reporting modest effects on p-tau concentrations and diagnostic accuracy [21–24]. The contribution of Aβ42/40 ratios to p-tau217 performance also varies across studies [14, 25]. Furthermore, physiological and vascular conditions may modulate plasma concentrations. p-tau levels tend to rise with age, although this may reflect increasing amyloid burden rather than ageing itself, and may also be influenced by renal function, with more pronounced effects reported from CKD stage 3 onward, while effects in mild impairment appear limited for amyloid prediction models [24, 26–29]. Magnetic resonance imaging (MRI) markers of small-vessel disease, such as white-matter hyperintensities or lobar microbleeds, have shown weak associations in some studies with higher p-tau217 and lower Aβ42/40, possibly reflecting mixed vascular–amyloid co-pathology such as cerebral amyloid angiopathy (CAA) [27, 30, 31]. Finally, although the cross-sectional diagnostic accuracy of plasma p-tau217 is well established, longitudinal studies indicate associations with faster cognitive decline and clinical progression, but its prognostic value in real-world settings remains to be determined [32–34].

In this study, we assessed the diagnostic performance of plasma p-tau217 measured on the Roche Elecsys platform in a memory-clinic cohort using CSF biomarkers as the reference standard. We benchmarked Elecsys p-tau217 against Fujirebio p-tau217 and Elecsys p-tau181, assessed the added value of APOE-ε4 carrier status and amyloid measures (Aβ42 and Aβ42/40 measured on Lumipulse), and explored the influence of clinical covariates. Finally, we evaluated whether baseline Elecsys p-tau217 is associated with longitudinal cognitive decline.

Materials and methods

Study population

Patients were selected from the Cognitive Centre Ghent University (CCUG)-Biobank, an ongoing prospective study initiated in 2021. The biobank collects blood and CSF samples, along with detailed clinical, neuropsychological, and imaging data from individuals referred for evaluation of cognitive complaints at the Cognitive

Centre of Ghent University Hospital. The cohort was predominantly of Belgian background. For the present study, we included consecutively enrolled participants up to the date of data extraction ($n = 187$) with available plasma, in whom the presence of underlying AD pathology was determined based on CSF biomarker results (CSF Aβ42/40 ratio, total tau, and p-tau181) ($n = 180$) or, in a minority of cases, amyloid-PET ($n = 7$). Based on these results, patients were categorized into those having AD ($n = 103$) or non-AD cognitive disorders ($n = 84$). The non-AD group consisted of patients with cognitive complaints attributed to non-AD neurodegenerative or other neurological disorders ($n = 54$), and patients with psychiatric or psychological causes of cognitive complaints ($n = 30$). In addition, 16 participants with discordant CSF AT(N) profiles were identified but excluded from these comparisons; these cases are described separately in the Results (Sect. 3.6).

Ethical approval

This study was approved by the local ethics committee of the Ghent University Hospital (ONZ-2024-0422). All procedures were conducted in accordance with the World Medical Association Declaration of Helsinki. Written informed consent was obtained from all participants before inclusion in the study. When decision-making capacity was lacking, consent was obtained from a legally authorised representative.

Blood collection and storage

Blood samples were collected in 4mL K2 EDTA tubes and subsequently centrifuged (2000*g* x 10 min, 4 °C) within 2 h after extraction. Next, plasma was aliquoted into polypropylene cryotubes and directly stored at -80 °C until analysis. All blood samples were collected, processed, and stored at – 80 °C in the CCUG-Biobank at Ghent University Hospital, following standardized procedures. On the day of analysis, the plasma samples were brought to room temperature. Following the manufacturer's instructions, plasma samples were centrifuged at 2000*g* for 5 min. The plasma was then transferred to the instrument cuvettes for analysis.

Biomarker measurements

Plasma concentrations of p-tau181 (Elecsys Phospho-Tau (181P) Plasma RUO; lot# 99065201), p-tau217 (Elecsys Phospho-Tau (217P) Plasma RUO, lot# 990760), and APOE-ε4 (Elecsys APOE-ε4 Plasma RUO, lot# 99065401) were measured using electrochemiluminescence immunoassays (ECLIA) on the cobas® pro e801 Elecsys immunochemistry analyser (Roche Diagnostics, Mannheim, Germany). For APOE-ε4, this plasma proteotyping assay yields ε4 carrier status (carrier vs. non-carrier). Plasma p-tau217 (Lumipulse G pTau217 Plasma RUO; lot# 5086), Aβ40 (Lumipulse G β-Amyloid 1-40-N Plasma RUO; V&V lot# 5), and Aβ42 (Lumipulse G β-Amyloid 1-42-N Plasma RUO; V&V lot# 8) concentrations were additionally measured using chemiluminescent enzyme immunoassays (CLEIA) on the LUMIPULSE G1200 platform (Fujirebio Europe). On the day of analysis, plasma samples were allowed to reach room temperature, thoroughly mixed, centrifuged at 2000*g* for 5 min, and then transferred into dedicated cuvettes for analysis on both platforms.

CSF biomarkers (Aβ42, Aβ42/40 ratio, p-tau181, and total tau) were measured on the LUMIPULSE G1200 platform (Fujirebio, Ghent, Belgium) in a blinded manner. The diagnosis of AD pathology was based on validated cut-off values according to the reference laboratory standards (Aβ42/Aβ40 ratio ≤ 0.070, Aβ42 ≤ 722 pg/mL, p-tau181 ≥ 50 pg/mL, and total tau ≥ 403 pg/mL) [35]. Plasma creatinine was determined using the Jaffe reaction on the cobas® pro c703 platform (Roche Diagnostics), and eGFR was calculated with the CKD-EPI 2009 equation. Plasma creatinine was measured on the same blood samples.

Statistical analysis

Baseline demographic and clinical characteristics were compared between the AD and non-AD groups using appropriate statistical tests (χ^2 for categorical variables and independent-samples t-tests or Mann–Whitney U tests for continuous variables, as appropriate). Estimated effect sizes (ES) are reported if relevant, considering 0.2 as a small effect, 0.5 as medium, and 0.8 as large.

The sample size (AD = 103, non-AD = 84) provides sufficient statistical power to detect differences in plasma p-tau levels between the two groups. Assuming a two-sided significance level of 0.05 and a medium to large, expected effect size (Cohen's $d = 0.5$-0,8), the calculated statistical power is approximately between 90% and 99%.

The precision of the p-tau181 and p-tau217 assays was determined using commercial quality control (QC) solutions (PreciControl pT181p Elecsys RUO and Phospho-Tau (217P) PreciControl Elecsys RUO, respectively) following the CLSI EP15-A3 protocol [36] with five measurements in a single run during five non-consecutive days. Linearity was measured by diluting a sample containing a high concentration (p-tau 181 and p-tau 217) according to the manufacturer's instructions. Each dilution was determined in triplicate, and the mean value was considered the correct result. The lower limit of quantitation (LLOQ) was determined using a predefined CV specification of 20% [35]. Samples with a measured concentration below the LLOQ were interpreted using the LLOQ value.

The diagnostic accuracy for distinguishing AD from non-AD cognitive disorders was assessed using the area under the ROC curve (AUC). Optimal diagnostic cut-off values for individual biomarkers and their combinations were identified by maximizing the Youden index. The added diagnostic value of APOE-ε4 carrier status was examined by including it as a binary covariate in logistic regression models predicting AD status. AUCs for different biomarkers and models were compared using the DeLong test.

In addition, a two-cut-off approach was applied, defining cut-offs at 95% sensitivity and 95% specificity to account for an intermediate biomarker zone. These cutoff values are presented alongside corresponding sensitivity, specificity, and likelihood ratios (calculated as sensitivity / (1 – specificity)).

Data are reported as medians with interquartile ranges (IQR). Quantitative comparisons between methods were conducted using Passing–Bablok regression and Bland–Altman analysis.

The impact of patient-related factors, such as age, sex, kidney function, and APOE-ε4 status, on biomarker concentrations were evaluated using Spearman rank correlation and, where appropriate, multivariable linear regression analysis.

Longitudinal associations between baseline plasma p-tau217 and Montreal Cognitive Assessment (MoCA) scores were evaluated using linear mixed-effects models with random intercepts and slopes, including an interaction between baseline p-tau217 and time to assess whether baseline p-tau217 is associated with the rate of MoCA change over follow-up, adjusted for age, sex, and baseline MoCA.

All statistical analyses were conducted using SPSS version 29.0, except for the Passing–Bablok and Bland–Altman analyses, which were performed with MedCalc (version 15.6.1., Mariakerke, Belgium). A p-value of <0.05 was considered statistically significant.

Table 1 Demographic, clinical and neurological characteristics of study cohort (AD vs. Non-AD)

Age, median (IQR, Q1, Q3)	AD ($n=103$)	Non-AD ($n=84$)	*p*-value	ES
	72 (10, 67, 77)	**70 (13, 63, 76)**	**0.132**	**0,11**
Male, n (%)	49%	64%	0.031	0,16
MoCA at baseline, median (IQR, Q1, Q3)	19 (7, 16.5, 23.5)	22 (9, 17.5, 26.5)	0.015	0,18
eGFR, median (IQR, Q1, Q3)	86 (22, 71, 93)	86,5 (25, 68, 93)	0.881	0,01
2 or more lobar microbleeds on MRI	9,8%	7,3%	0.552	0,04
Fazekas grade 3, n (%)	2,9%	4,7%	0.508	0,05
APOE ε4 carrier status, n (%)	74.7%	13.1%	< 0.001	0.61
Elecsys p-tau181, median (IQR, Q1, Q3)	1.600 (0.710, 1.370, 2.080)	0.855 (0.350, 0.690, 1.040)	< 0.001	0,69
Elecsys p-tau217, median (IQR, Q1, Q3)	0.791 (0.499, 0.581, 1.080)	0.242 (0.094, 0.199, 0.293)	< 0.001	0,75
Lumipulse p-tau217, median (IQR, Q1, Q3)	0.495 (0.453, 0.312, 0.765)	0.094 (0.054, 0.073, 0.127)	< 0.001	0,77
Lumipulse Aβ42, median (IQR, Q1, Q3)	16.3 (3.8, 14.4, 18.2)	21.3 (5.3, 18.2, 23.5)	< 0.001	0,59
Lumipulse Aβ40, median (IQR, Q1, Q3)	250.4 (41.9, 230.9, 272.8)	264.0 (77.5, 228.2, 305.7)	0.037	0,15
Lumipulse p-tau217 / Aβ42 (IQR, Q1, Q3)	0.03 (0.033, 0.019, 0.052)	0.04 (0.02, 0.04, 0.06)	< 0.001	0.78
Lumipulse p-tau217 / (Aβ42/ Aβ40) (IQR, Q1, Q3)	7.642 (7.64, 4.856, 11.904)	1,199 (0.78, 0.871, 1.650)	< 0,001	0,77

IQR Interquartile range, *ES* Estimated effect size, *MOCA* Montreal Cognitive Assessment

Results

Description of study population

The study included 187 patients with cognitive complaints referred to the Cognitive Centre, Ghent University Hospital comprising 103 with AD and 84 with non-AD cognitive disorders (AD prevalence 55.1%). Classification was based on a CSF biomarker reference standard, with amyloid-PET used in a minority of patients. Baseline demographic and clinical characteristics are summarized in Table 1.

There was no significant age difference between the AD and non-AD group (median [IQR] 72 [67–77] years vs. 70 [63–76] years; $p=0.132$). The proportion of men was lower in the AD group (49% vs. 64%; $p=0.031$). Median estimated glomerular filtration rate (eGFR) values were comparable between groups (86 [71–93] vs. 86.5 [68–93] mL/min/1.73 m^2; $p=0.881$). Cognitive performance at baseline, as measured by the MoCA, was lower in the AD group (median [IQR] 19 [16.5–23.5]) than in the non-AD group (22 [17.5–26.5]; $p=0.015$). MRI markers of small vessel disease showed similar distributions across groups, with two or more lobar microbleeds observed in 9.8% of AD and 7.3% of non-AD patients ($p=0.552$) and severe white-matter hyperintensities (Fazekas grade 3) in 2.9% and 4.7%, respectively ($p=0.508$). There was a significant difference between the AD and non-AD groups for the plasma biomarkers, with higher p-tau concentrations and lower amyloid-β levels observed in the AD group.

Within the non-AD group, cognitive disorders comprised non-AD neurological disorders (ND, $n=54$), including other neurodegenerative and vascular aetiologies, and psychiatric or psychological causes of cognitive complaints (PSY, $n=30$). The diagnostic composition and clinical characteristics of the ND and PSY subgroups are provided in the Supplementary file. Briefly, ND patients were older and had lower baseline MoCA scores than PSY patients (age 71.5 [65.0–77.25] vs. 66.0 [61.75–72.25] years, $p = 0.012$; MoCA 19.5 [15.25–23.75] vs. 26.0 [23.0–27.0], $p < 0.001$), and baseline plasma biomarker distributions differed modestly between subgroups (Supplementary Table 2).

Analytical performance

Precision, linearity and limit of quantification

Imprecision was estimated using commercially available QC solutions (PreciControl pT181p Elecsys RUO and Phospho-Tau (217P) PreciControl Elecsys RUO, Roche Diagnostics, Mannheim, Germany). Depending on the concentration, imprecision for p-tau181 ranged from 0.5 to 3.8%, whereas for p-tau217 it ranged from 0.5 to 8.6% (Supplemental Table 1). Linearity was maintained down to 0.30 pg/mL and 0.12 pg/mL for p-tau181 and p-tau217, respectively (Supplementary Figure S1). The LLOQ was estimated at 0.32 pg/mL for p-tau181 and 0.10 pg/mL for p-tau217.

Method comparison

Passing–Bablok regression analysis demonstrated both proportional (slope = 1.363; 95% CI: 1.251–1.484) and systematic (intercept = 0.113; 95% CI: 0.098–0.133) differences in p-tau217 concentrations between the two assays (Fig. 1A). A statistically significant correlation was observed between the assays ($r = 0.88$, $p < 0.0001$). Furthermore, Bland–Altman analysis, performed by plotting the differences between measurements against their means, indicated discrepancies of up to 59% between the assays (Fig. 1B). Five out of the 187 included samples had a concentration below the LOQ (0.10 pg/mL).

Diagnostic accuracy of Elecsys p-tau217 compared with Lumipulse p-tau217 and Elecsys p-tau181

ROC analyses showed good discrimination between AD and non-AD cases, with AUCs of 0.939 (95% CI 0.895–0.969) for Elecsys p-tau217, 0.950 (95% CI 0.907–0.976) for Lumipulse p-tau217, and 0.903 (95% CI 0.852–0.942) for Elecsys p-tau181 (Table 2; Fig. 2). All assays achieved Youden-derived cut-offs with both sensitivity and specificity ≥ 83% [35], with Elecsys p-tau217 providing the highest sensitivity (92.2%) and Lumipulse p-tau217 providing the highest specificity (96.2%). Optimal cut-off values were 0.371 pg/mL for Elecsys p-tau217, 1.150 pg/mL for Elecsys p-tau181, and 0.246 pg/mL for Lumipulse p-tau217. The DeLong test showed comparable discriminatory performance between the two p-tau217 assays ($p = 0.485$) while the AUC of Elecsys p-tau181 was significantly lower than that of Elecsys p-tau217 ($p = 0.043$).

To further refine diagnostic classification, a two-cut-off approach was applied to define a grey zone separating clearly normal and abnormal results (Table 3). For Elecsys p-tau217, the upper cut-off (95% specificity) was 0.497 pg/mL and the lower cut-off (95% sensitivity) was 0.275 pg/mL. This range encompassed 19.9% (37/186) of patients, of whom 32.4% (12/37) were AD and 67.6% (25/37) non-AD (Fig. 3**)**. For Lumipulse p-tau217, the upper cut-off was 0.246 pg/mL and the lower cut-off 0.130 pg/mL, defining a grey zone including 11.9% (22/186) of patients, of whom 45.5% (10/22) were AD and 54.5% (12/22) non-AD. For Elecsys p-tau181, the upper cut-off was 1.505 pg/mL and the lower cut-off 0.936 pg/mL, defining a grey zone including 33.2% (62/187) of

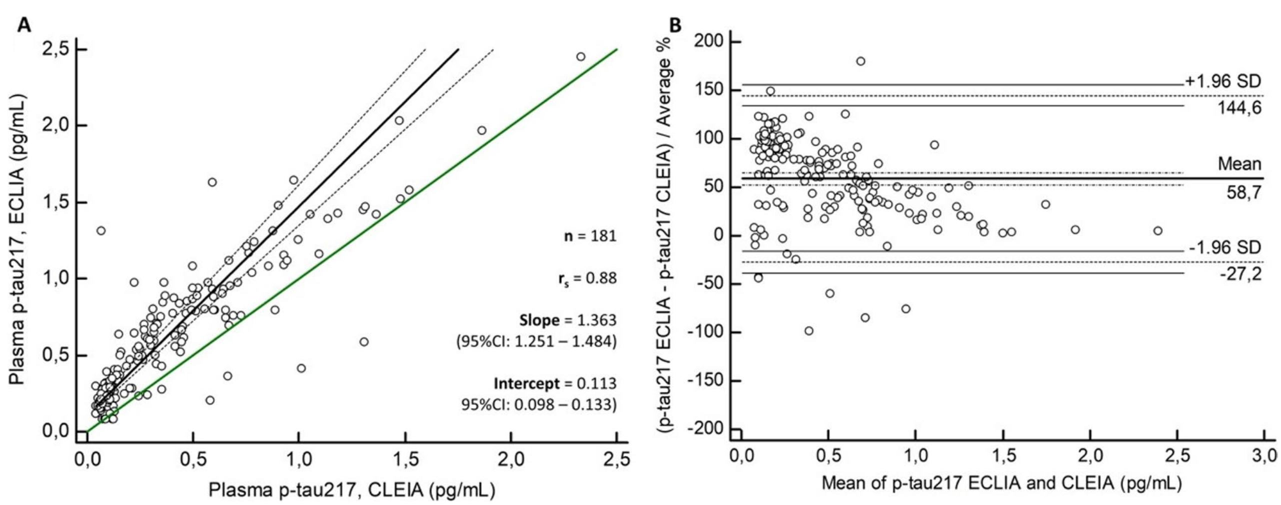

Fig. 1 Comparison of p-tau217 between ECLIA (Roche; Elecsys) and CLEIA (Fujirebio; Lumipulse). Passing-Bablok regression analysis (1). The bold line shows the fitted regression line. The green line indicates the line of identify. The 95%CI of the regression curve is shown as small, dashed lines. The regression equation (slope and intercept with corresponding 95%CI) and the Pearson correlation coefficient (r_s) are indicated. Bland-Altman plot demonstrating the relative difference obtained with ECLIA (Roche) against CLEIA (Fujirebio) (2). Horizontal lines are drawn at the mean difference between the two methods, and at the limits of agreement, which are defined as the mean difference plus and minus 1.96 times the standard deviation of the differences. The dotted horizontal lines present the 95% confidence intervals around the mean difference and the upper and lower limits of agreement. Abbreviations: ECLIA: electrochemiluminescence immunoassays; CLEIA: Chemiluminescence enzyme immunoassay

Table 2 Diagnostic performance using single cut-off approach

Parameter	*n*	AUC (95%CI)	Optimal cut-off value (pg/mL)	Sensitivity (95% CI)	Specificity (95% CI)	Accuracies (95% CI)	PPV (95% CI)	NPV (95% CI)	TP / TN
Elecsys p-tau217	186	0.939 (0.895–0.969)	0.371	92.2% (85.3–96.6%)	89.2% (80.4–94.9%)	91% (87–95%)	91.3% (84.1–95,3%)	90.4% (82,1–94.7%)	95 / 75
Elecsys p-tau181	187	0.903 (0.852 0.942)	1.150	85.4% (77.1–91.6%)	86.9% (77.8–93.3%)	86% (81–91%)	88.9% (91.0–94.3%)	93.0% (73.4–90.1%)	88 / 73
Lumipulse p-tau217	186	0.950 (0.907–0.976)	0.246	85.4% (77.1–91.6%)	96.2% (89.3–99.2%)	88% (83–92%)	93,6% (86,7–97,0%)	83.8% (75,1–90,0%)	86 / 78
Lumipulse p-tau217/Aβ42	182	0.957 (0.916–0.981)	0.0115	89.3% (81.7–94.5%)	94.9% (87.5–98.6%)	95% (80–98%)	95.8% (89.7–98.9%)	87.2% (78.3–93.4%)	92 / 80
Lumipulse p-tau217/(Aβ42/Aβ40)	182	0.957 (0.917–0.982)	3.891	85.4% (77.1–91.6%)	96.2% (89.3–99.2%)	92% (87–96%)	96.7% (90.7–99.3%)	83.5% (74.3–90.5%)	88 / 80
Elecsys p-tau181 + APOE-ε4	187	0.943 (0.907–0.978)	N.A.	96.1% (91.2–99.4%)	85.2% (76.0–92.0%)	91% (87–95%)	88.4% (83.1–94.2%)	94.7% (90.2–99.6%)	99 / 75
Elecsys p-tau217 + APOE-ε4	187	0.970 (0.943–0.992)	N.A.	90.2% (83.2–95.2%)	92.7% (87.1–98.3%)	91% (87–95%)	92.9% (88.2–97.6%)	93.9% (88.0–98.8%)	92 / 78
Lumipulse p-tau217 + APOE-ε4	186	0.959 (0.928–0.989)	N.A.	95.1% (89.1–96.4%)	91.6% (83.7–95.6%	93% (88–96%)	92.4% (85.7–96.1%	90,2% (84.9–96.6%)	97 / 76
Lumipulse (p-tau217 / Aβ42) + APOE-ε4	182	0.965 (0.938–0.993)	N.A.	94.2% (87.9–96.4%)	92.4% (84.7–95.6%)	96% (92–98%)	94.2% (87.9–96.4%)	92.9% (89.3–96.2%)	97 / 78

Abbreviations: *CI* Confidence interval, *AUC* Area under the ROC curve, *ROC* Receiver operating characteristics curve, *N.A.* Not applicable

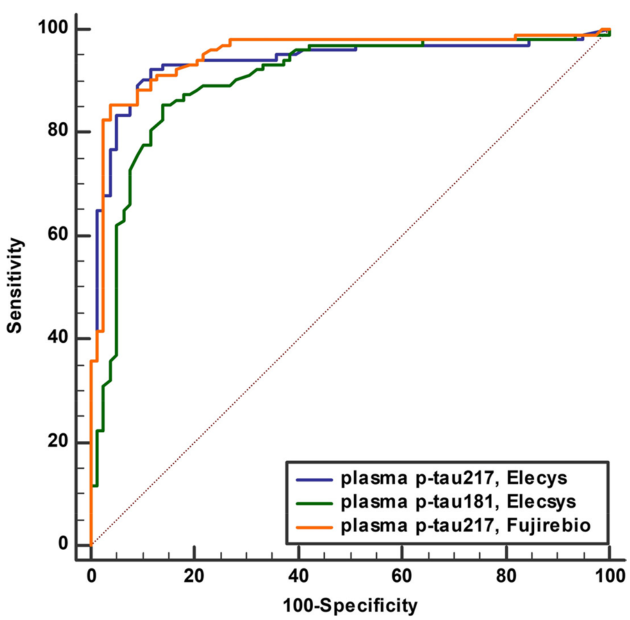

Fig. 2 ROC curves demonstrating the diagnostic performance of Elecsys p-tau 217, Lumipulse p-tau 217 and Elecsys p-tau 181

patients, of whom 54.8% (34/62) were AD and 45.2% (28/62) non-AD.

Given the heterogeneous composition of the non-AD group, we performed subgroup analyses comparing AD with the ND and PSY groups (Supplementary Tables 3 and 4). Using the single cut-off approach, AUCs for Elecsys p-tau217, Lumipulse p-tau217 and Lumipulse p-tau217/Aβ42 ranged from 0.932 to 0.947 for AD vs. ND and from 0.953 to 0.970 for AD vs. PSY, consistent with the overall AD vs. non-AD analysis (Table 2). Using the two-cut-off approach, intermediate rates differed by subgroup and biomarker: in AD vs. ND, intermediates were 31.8% for Elecsys p-tau217 and 26.1% for Lumipulse p-tau217, while Lumipulse p-tau217/Aβ42 yielded 3.8% intermediates; in AD vs. PSY, intermediates were 5–6% for p-tau217 and <1% for the p-tau217/Aβ42 ratio (Supplementary Tables 3 and 4). Using this two-cut-off approach, sensitivity and specificity were ≥90% for Elecsys p-tau217, Lumipulse p-tau217 and Lumipulse p-tau217/Aβ42.

Diagnostic value of plasma APOE ε4 carrier status

To explore the added diagnostic value of APOE ε4 carrier status, we examined its contribution in combination with plasma biomarker measures. In logistic regression models, adding APOE ε4 increased discriminative performance for Elecsys p-tau181 (AUC 0.903 to 0.943; $p = 0.010$) and for Elecsys p-tau217 (AUC 0.939 to 0.970; $p = 0.02$), while AUC changes were more modest for Lumipulse p-tau217 (AUC 0.950 to 0.959; $p = 0.101$) and Lumipulse p-tau217/Aβ42 (AUC 0.957 to 0.965; $p = 0.121$). In the two-cutoff approach, the proportion of patients classified in the intermediate zone decreased when APOE ε4 was incorporated (Elecsys p-tau181 from 33.2% to 18.2%, Elecsys p-tau217 from 20.0% to 11.0%, Lumipulse p-tau217 from 11.9% to 8.7%, and Lumipulse p-tau217/Aβ42 from 10.0% to 8.7%). Across these combined models, positive and negative predictive values were 92.4% or higher.

Subgroup analyses for p-tau217 showed that APOE-ε4 carriers had higher PPV (95–100%) but lower NPV (53–60%), whereas non-carriers showed the opposite pattern,

Table 3 Diagnostic performance using two cut-off approach (95% sensitivity/specificity thresholds)

Parameter	Lower cut-off (pg/mL)	Upper cut-off (pg/mL)	Accuracy (95% CI)	PPV (95% CI)	NPV (95% CI)	TP / TN	% intermediates
Elecsys *p*-tau217	0.275	0.497	94% (90–98%)	95.6% (89.6–98.2%)	91.5% (81.8–96.1%)	86 / 54	19.9%
Elecsys p-tau181	0.936	1.505	93% (88–97%)	94.1% (85.8–97.6%)	91.2% (80.1–96.3%)	64 / 52	33.2%
Lumipulse p-tau217	0.130	0.246	93% (89–97%)	93.6% (86.7–97.0%)	92.8% (84.3–96.9%)	88 / 65	11.9%
Lumipulse p-tau217/Aβ42	0.0071	0.0139	95% (91–98%)	95.7% (89.6–98.6%)	93.0% (84.6–97.3%)	89 / 66	10.0%
Lumipulse p-tau217/ (Aβ42/Aβ40)	0.912	3.841	90% (85–94%)	95.7% (89.4–98.6%)	81.7% (71.1–89.4%)	88 / 58	10.5%
Elecsys p-tau217 + APOE-ε4	N.A.	N.A.	94% (90–98%)	94.4% (87.5–97.9%)	93.4% (85.5–97.3%)	85 / 71	11.0%
Elecsys p-tau181 + APOE-ε4	N.A.	N.A.	93% (90–98%)	93.4% (87.8–99.0%)	93.4% (87.8–99.0%)	71 / 71	18.2%
Lumipulse p-tau217 + APOE-ε4	N.A.	N.A.	95% (89–96%)	92.4% (85.7–96.1%)	92.7% (84.9–96.6%)	83 / 76	8.7%
Lumipulse (p-tau217 / Aβ42) + APOE-ε4	N.A.	N.A.	95% (90–97%)	95.7% (89.5–96.1%)	93.2% (85.1–95.1%)	90 / 69	8.7%

Abbreviations: *CI* Confidence interval, *AUC* Area under the ROC curve, *ROC* Receiver operating characteristics curve, *N.A.* Not applicable

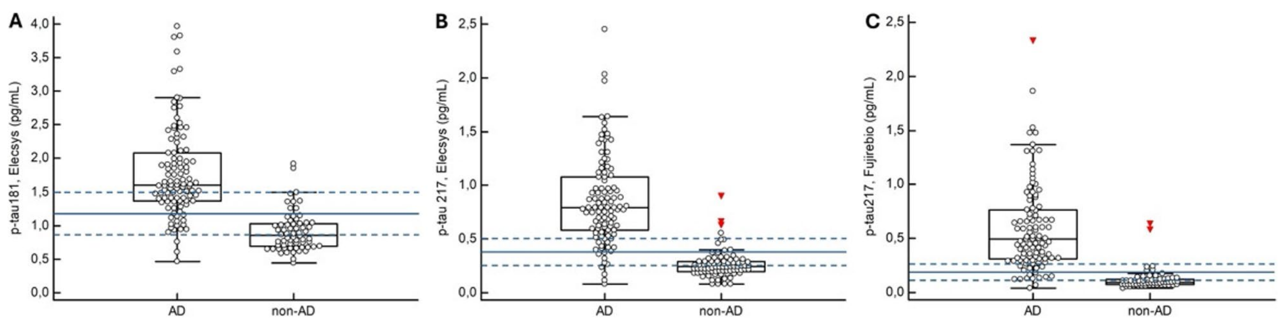

Fig. 3 Distribution of plasma Elecsys p-tau181 (**A**), Elecsys p-tau217 (**B**), and Fujirebio Lumipulse p-tau217 (**C**) in AD and non-AD patients. For each biomarker, the optimal single cut-off is indicated by the solid blue line; dashed lines indicate thresholds corresponding to 95% sensitivity (lower dashed line) and 95% specificity (upper dashed line). Far-out values, defined as values below Q1 minus 3×IQR or above Q3 plus 3×IQR, are indicated by red triangles

with NPV up to 98% and PPV around 80–85%. These findings indicate that APOE-ε4 primarily reduces diagnostic uncertainty by shifting the interpretation of intermediate p-tau217 results toward AD positivity in carriers and toward AD negativity in non-carriers.

Adjustment with Aβ42 or the Aβ42/40 ratio

Within the Fujirebio platform, where Aβ42 and Aβ40 measurements were available, combining plasma p-tau217 with Aβ42 and Aβ42/40 ratio did not result in a statistically significantly increase the AUC (0.950 vs. 0.957; $p = 0.058$ and $p = 0.322$, respectively). The proportion of intermediate cases, PPV and NPV remained unchanged (Tables 2 and 3), while accuracy increased (95% and 92% vs. 88%). Comparing Lumipulse p-tau217/Aβ42 and Elecsys p-tau217, there was no significant difference in diagnostic performance (AUC 0.957 vs. 0.939; $p = 0.320$), with Lumipulse p-tau/ Aβ42 having a slightly higher positive predictive value (95.8% vs. 91.3%) and slightly lower negative predictive value (87.2% vs. 90.4%).

Discordant CSF biomarker profiles

Discordant CSF profiles were excluded from the primary analyses and were defined as amyloid abnormal without CSF tau abnormality (A+T−N−, potentially reflecting an early amyloid-positive stage and/or mixed pathology; $n = 15$) or other discordant AT(N) profiles ($n = 1$). Overall, discordant cases differed clearly from the AD group across plasma p-tau measures ($p < 0.001$). Compared with the non-AD group, Elecsys p-tau181 and Elecsys p-tau217 levels were similar, while small differences were observed for Lumipulse p-tau217 ($p = 0.039$). Applying the Elecsys p-tau217 single cut-off from the primary AD versus non-AD analysis, 62.5% discordant cases were below and 37.5% were above the threshold, indicating substantial heterogeneity within this subgroup. In sensitivity analyses, classifying discordant cases as AD resulted in only modest changes in AUC, sensitivity and specificity and did not materially alter the optimal single cut-off values (Supplementary Table 5). Overall, these exploratory findings indicate substantial heterogeneity among discordant CSF profiles, with elevated plasma p-tau217 in a subset despite normal CSF tau cut-offs and support the robustness of the main diagnostic cut-offs.

Influence of clinical cofactors on Elecsys p-tau217

Linear regression analyses were performed to evaluate the impact of clinical cofactors on baseline Elecsys p-tau217 levels. Significant associations were observed with diagnostic group and baseline MoCA score

($p < 0.001$, β=-0.267), consistent with higher p-tau217 concentrations in patients with AD and in those with lower cognitive performance. In contrast, no significant associations were found with age ($p = 0.449$, β = 0.064), sex ($p = 0.054$, β = 0.154), renal function (eGFR) ($p = 0.918$, β=-0.009), severe white-matter hyperintensities (Fazekas score = 3) ($p = 0.072$, β = 0.142), or the presence of two or more strictly lobar microbleeds indicative of CAA ($p = 0.308$, β = 0.079). Twenty-one patients had an eGFR below 60 mL/min/1.73 m² in our cohort, consistent with chronic kidney disease (CKD stage 3), and none had an eGFR below 30 mL/min/1.73 m² (CKD stage 4–5). Plasma Elecsys p-tau217 did not differ significantly across eGFR categories in the overall cohort ($p = 0.07$).

Baseline MoCA scores were significantly lower in the AD group than in the non-AD group, raising the possibility of confounding by cognitive severity. Therefore, we performed MoCA-adjusted logistic regression analyses for each plasma biomarker. After MoCA adjustment, higher levels of Elecsys p-tau181 (OR 1.49, 95% CI 1.32–1.69, $p < 0.001$), Elecsys p-tau217 (OR 3.33, 95% CI 2.22–5.00, $p < 0.001$) and Lumipulse p-tau217 (OR 4.55, 95% CI 2.78–7.69, $p < 0.001$), and lower levels of Aβ42 (OR 0.96, 95% CI 0.95–0.98, $p < 0.001$) remained associated with AD diagnosis. MoCA was not independently associated with diagnosis in the p-tau analyses, whereas it remained independently associated in the amyloid analyses (Aβ42: OR 0.93, 95% CI 0.85–1.00, $p = 0.045$; Aβ40: OR 0.92, 95% CI 0.85–0.99, $p = 0.019$).

Cognitive decline and p-tau217

Among AD patients with longitudinal follow-up ($n = 69$; median follow up 2.1 years), higher baseline plasma Elecsys p-tau217 levels were associated with lower MoCA scores at baseline and showed a trend toward faster cognitive decline over time ($p = 0.07$) in models adjusted for age and sex.

Discussion

In this study, we provide one of the first comprehensive evaluations of plasma p-tau217 measured on the fully automated Roche Elecsys platform in a large real-world memory-clinic cohort. Our results demonstrate that Elecsys p-tau217 shows excellent diagnostic accuracy for distinguishing AD from non-AD cognitive disorders (AUC 0.939), outperforming Elecsys p-tau181 (AUC 0.903; $p = 0.043$) and showing comparable performance to the Fujirebio Lumipulse p-tau217 assay (AUC 0.950; $p = 0.485$). A recent meta-analysis reported a pooled AUC of 0.911 (95% CI 0.889–0.924) for p-tau217 and 0.815 (95% CI 0.802–0.829) for p-tau181 in cognitively impaired patients across a broad range of plasma p-tau assays. Notably, Elecsys p-tau217 was not included [14]. Much of the existing evidence comes from research assays and automated immunoassay platforms, including Fujirebio Lumipulse based testing in clinical care settings [24, 37]. A recent head-to-head comparison showed consistently high diagnostic performance across multiple plasma p-tau217 assays, including immunoassays (ALZpath, Lilly, and Janssen/J&J) and a Washington University mass spectrometry assay, for detecting amyloid PET positivity (AUC 0.91 to 0.96), with classification accuracies of 83% to 93%, sensitivities of 84% to 91% and specificities of 85% to 94% [38]. Our findings build on these observations by showing that comparable performance can be achieved on a fully automated, high-throughput analyser platform used in routine laboratory workflows, with an AUC of 0.939 (sensitivity 92%, specificity 89%) for Elecsys p-tau217. Importantly, implementation on such platforms facilitates routine clinical use by streamlining assay execution, reducing operator-dependent variability, and enabling high-throughput testing within existing laboratory workflows.

While plasma p-tau217 distinguished between the AD and non-AD groups, absolute concentrations differed between the evaluated assays, consistent with prior reports [39–41]. Differences in sample matrix (for example serum versus plasma) may also contribute to variability in measured concentrations [42, 43]. The development of certified reference materials would be an important step toward assay standardization, improved calibration and analytical sensitivity, and reduced fixed and proportional bias [39]. Such harmonization is essential to ensure accurate and reproducible results across laboratories and platforms and to enable reliable interpretation and comparison of biomarker concentrations across studies.

The blood based biomarker (BBM) workgroup suggested that the minimal acceptable performance for blood biomarkers as a confirmatory test should be equivalent to CSF tests with a sensitivity and specificity of 90% or higher [44]. In our cohort, Elecsys p-tau217 nearly met these criteria using a single cutoff (sensitivity 92.2%, specificity 89.2%). To further optimize diagnostic performance, we applied a two-cut-off approach defining an intermediate range. Using thresholds corresponding to 95% sensitivity and specificity resulted in 19.9% intermediate results, within the range of 15–20% as recommended by the BBM workgroup. For Lumipulse p-tau217, using a 95% sensitivity/specificity threshold showed 11.9% intermediates. These findings illustrate the trade-off between diagnostic stringency and the number of intermediate results, underscoring that cut-off selection should balance false positives against indeterminate outcomes, particularly in the context of high-cost disease-modifying therapies. Patients with intermediate values should undergo confirmatory testing using CSF or amyloid-PET to establish the underlying pathology.

The importance of defining clinically meaningful thresholds is exemplified by the recent FDA approval of the Elecsys Phospho-Tau 181 (p-tau181) assay as an aid for evaluating cognitive complaints in adults aged ≥ 55 years undergoing assessment for AD in primary care settings. The FDA-approved cut-off (0.934 pg/mL) differs notably from our study threshold (1.150 pg/mL), reflecting fundamental differences in intended use. As a rule-out test, the FDA-cleared assay prioritizes sensitivity (91.0%) over specificity (69.8%), aiming to minimize false negatives in a screening context. Similarly, Hibar et al. [20] positioned Elecsys p-tau217 as a prescreening tool for amyloid-beta pathology, selecting a low threshold (p-tau217 < 0.189 pg/mL) that maximized sensitivity (95.5–98.6%) and NPV at the cost of low specificity (29.2–50.7%) in both cognitively unimpaired and impaired individuals.

In contrast, our memory-clinic cohort required thresholds with confirmatory intent for cognitively impaired patients, where higher specificity is essential to support diagnostic confirmation and inform therapeutic decision-making. Applying a higher cutoff (< 0.371 pg/mL) for p-tau217, we achieved a more balanced diagnostic performance with 96% sensitivity and 88% specificity. These findings underscore that threshold selection must align with clinical context and the relative consequences of false-positive versus false-negative results. While Elecsys p-tau181 offers an FDA-cleared option for primary care screening, p-tau217 may serve as the preferred confirmatory blood-based marker when available, consistent with BBM workgroup recommendations targeting ≥ 90% sensitivity and specificity for confirmatory biomarkers [44, 45].

Given the heterogeneous composition of the non-AD group, we performed subgroup analyses and found broadly similar AUCs for AD versus both non-AD neurological and psychiatric/psychological subgroups. However, intermediate rates were higher in the non-AD neurological subgroup, suggesting that confirmatory CSF or amyloid-PET may be more frequently be required in this group.

The analytical evaluation of the cobas® pro Elecsys p-tau181 and p-tau217 assays revealed coefficients of variation (CVs) ranging from 0.5 to 3.8% and 0.5 to 8.6%, respectively. These values are consistent with precision estimates reported for other immunoassay platforms, including SIMOA, MSD, and CLEIA [27, 37, 46]. Method-comparison analyses showed that Elecsys and Lumipulse yielded different absolute p-tau217 concentrations but were highly correlated. This aligns with a Swedish head-to-head comparison of the Janssen/J&J SIMOA p-tau217 and Lilly MSD p-tau217 assays, which reported strong correlations (Spearman ρ = 0.69–0.71) and good agreement on Bland–Altman analyses [47]. Similarly, the Janssen/J&J SIMOA p-tau217 assay correlated closely with the ALZpath p-tau217 assay (Spearman ρ = 0.85), with > 95% individual-level agreement [48]. Together, these findings indicate that across cohorts and assay platforms, plasma p-tau217 measurements are strongly correlated, show high agreement, and provide comparable diagnostic performance.

Our findings further suggest that incorporating APOE-ε4 status with plasma p-tau217 improves overall discrimination and reduces the proportion of intermediate results. Adding APOE-ε4 to Elecsys p-tau217 increased the AUC from 0.939 to 0.970 ($p = 0.02$) and decreased the proportion of grey-zone results from 20% to 11%. In carriers of the ε4 allele, positive predictive values reached 95–100%, whereas non-carriers showed the highest negative predictive values (up to 98%). These findings indicate that integrating APOE carrier status, determined from blood, with plasma biomarkers could enhance clinical interpretation by refining the classification of borderline results. Plasma-based APOE-ε4 proteotyping has shown high agreement with genotyping in recent studies, supporting its use for risk stratification in clinical workflows [49]. In practice, this approach may help avoid unnecessary confirmatory CSF or PET investigations in ε4 carriers with an amnestic presentation, while maintaining a cautious approach for non-carriers in the same range. Broader validation and careful integration into clinical workflows will be required before routine implementation. Particular attention should be given to ethical and communication aspects surrounding disclosure of genetic-risk information, even when carrier status is inferred from blood-based assays rather than direct genotyping.

Combining plasma Lumipulse p-tau217 with Aβ42 or the Aβ42/Aβ40 ratio has been proposed to improve diagnostic performance [25]. In May 2025, the Lumipulse plasma p-tau217/Aβ42 assay became the first FDA-cleared blood biomarker test to aid the diagnosis of AD in symptomatic patients aged ≥ 55 years and showed high agreement with CSF biomarkers or amyloid PET in the pivotal study [50]. In our cohort measured on the Lumipulse platform, adding Aβ42 or the Aβ42/Aβ40 ratio to p-tau217 slightly reduced the proportion of intermediate results (11.9% vs. 10.0% vs. 10.5%) without a meaningful gain in overall discrimination (AUC 0.950 vs. 0.957 vs. 0.957). Diagnostic performance of Lumipulse p-tau217/Aβ42 was comparable to Elecsys p-tau217 in our cohort. Sensitivity and specificity were 89.3% and 94.9% for p-tau217/Aβ42 and 85.4% and 96.2% for p-tau217/(Aβ42/Aβ40), broadly in line with the performance reported in the FDA-cleared setting. In our cohort, p-tau217/Aβ42 and p-tau217/(Aβ42/Aβ40) performed similarly, however, the incremental value of Aβ42/Aβ40 may differ by cohort and analytical platform [51]. Overall, adding Aβ42 or Aβ42/Aβ40 may help classify borderline cases depending on the cohort, although the incremental benefit over p-tau217 alone was small in our cohort.

Discordant CSF ATN profiles, particularly A + T−N−, are often interpreted as an early Alzheimer continuum stage in which amyloid pathology is present while CSF tau biomarkers remain below established cut-offs. In our cohort, such CSF profiles, while on average similar to the non-AD group, showed substantial heterogeneity, with elevated plasma p-tau217 in a subset despite normal CSF p-tau181 and total tau. This pattern is compatible with published trajectory data suggesting that plasma p-tau217 can become abnormal at relatively low amyloid burden and may signal emerging AD-related tau pathophysiology before conventional CSF tau cut-offs are crossed in some individuals [52]. Alternative contributors to plasma and CSF discordance include epitope and assay differences (p-tau217 versus p-tau181), cut-off dependence and preanalytical variability, and, in selected cases, peripheral influences on plasma tau measures. Clinically, these observations underscore that an isolated elevation of plasma p-tau217 should be interpreted in the context of phenotype and other AD indicators, and that confirmatory CSF or amyloid-PET should be considered when results are discordant or when the clinical picture is atypical.

Physiological and vascular factors have been proposed to influence plasma p-tau levels, including ageing, reduced renal clearance, and small-vessel disease [24, 27, 28]. In our memory-clinic cohort, with a median age of approximately 70 years, Elecsys p-tau217 concentrations were not significantly affected by age, sex, renal function, severe white-matter hyperintensities, or the presence of two or more strictly lobar microbleeds, which may indicate concomitant CAA. The absence of an association between p-tau217 and eGFR in our study may therefore reflect the relatively small number of participants with impaired renal function and the absence of cases with advanced CKD, in whom stronger effects have been reported [26, 29]. Recent studies have indicated that using p-tau217/total tau ratios, as determined by mass spectrometry–based assays, may mitigate the impact of kidney dysfunction on plasma p-tau217, but such normalized ratios are not currently available for the Elecsys and Lumipulse immunoassays and were therefore not evaluated here [53]. Overall, these findings suggest that p-tau217 values primarily reflect disease-related rather than physiological variation in typical memory-clinic populations.

Higher plasma p-tau217 levels were associated with poorer baseline cognition, and baseline p-tau217 showed a trend toward faster MoCA decline over time in our cohort. While this longitudinal association did not reach statistical significance, the direction of effect is consistent with emerging longitudinal evidence linking higher p-tau217 to subsequent cognitive worsening and increasing tau burden over time [31–33]. Importantly, our longitudinal outcome was limited to MoCA, which has limited sensitivity for subtle within-subject change and may show floor effects in AD. More sensitive repeated neuropsychological measures were not available in a sufficiently large subset for robust modelling. Therefore, these longitudinal findings should be viewed as exploratory, and larger studies with repeated sensitive cognitive assessments are needed to clarify the strength and clinical relevance of this association.

Strengths and limitations

The main strengths of this study are the relatively large, well-characterised memory-clinic cohort with a CSF-based reference standard, and the direct head-to-head comparison across assays supported by method-comparison analyses (Passing Bablok and Bland Altman). We also report clinically interpretable measures such as grey-zone proportions and evaluate the added value of integrating APOE ε4 status in diagnostic models. Limitations include the single-centre design, the lack of cross-validation in an independent cohort, and a more extensive analytical evaluation in line with international validation criteria (e.g., interference testing, multi-lot and multi-operator precision, and assessment of the upper limit of quantification) was not performed, which was beyond the scope of this clinical performance study. Finally, the longitudinal follow-up subset was modest and relied on MoCA, so longitudinal findings should be considered exploratory.

Conclusion

Together, these findings highlight the potential of Elecsys p-tau217 as a scalable and clinically applicable biomarker for AD, with high diagnostic accuracy and possible associations with cognitive severity and longitudinal cognitive change. Future work should focus on multicentre validation in larger and more diverse populations, the harmonisation of assay-specific thresholds, and the development of longitudinal frameworks for clinical implementation.

Abbreviations

AD	Alzheimer's disease
AUC	Area under the ROC curve
CAA	Cerebral amyloid angiopathy
CSF	Cerebrospinal fluid
PET	Positron emission tomography
IQR	Interquartile range
MOCA	Montreal Cognitive Assessment
eGFR	Estimated glomerular filtration rate
MRI	Magnetic resonance imaging
ROC	Receiver operating characteristic
ECLIA	Electrochemiluminescence immunoassays
PPV	Positive predictive value
NPV	Negative predictive value
CI	Confidence interval

ES	Estimated effect size
CKD	Chronic kidney disease
CCUG	Cognitive Centre Ghent University
GFAP	Glial fibrillary acidic protein
NfL	Neurofilament light chain
QC	Quality control
LoQ	Limit of Quantification

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13195-026-01996-8.

Supplementary Material 1

Acknowledgements

Not applicable.

Authors' contributions

Jochen Titeca performed the main statistical analyses, data visualization, and figure preparation, and contributed to data interpretation and manuscript writing.Marta Scarioni contributed to patient selection, clinical assessment, and inclusion of participants in the Cognitive Centre Ghent University (CCUG) biobank and critically reviewed the manuscript.Matthijs Oyaert supervised and performed biomarker analyses, including sample processing and quality control of the Elecsys and Lumipulse assays, contributed to the statistical analyses, data visualization, and study design, and co-wrote and revised the manuscript.Tim Van Langenhove conceived the study design, supervised the overall project, contributed to the statistical analyses and data interpretation, and co-wrote and revised the manuscript.

Funding

Roche and Fujirebio provided the biomarker testing tools free of charge for this study. The companies had no role in study design, data collection, data analysis, interpretation, or manuscript preparation.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the local ethics committee of the Ghent University (ONZ-2024-0422). All procedures were conducted in accordance with the World Medical Association Declaration of Helsinki. Written informed consent was obtained from all participants before inclusion in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 6 November 2025 / Accepted: 22 February 2026
Published online: 28 February 2026

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